IWS-RL: understanding the project (edition 3, morning of 2 Nov 2026)

Repo: https://github.com/adithyx-0/iws-rl | Plan for the next 30 hours: `docs/PLAN_v1.md` | Full history: `docs/PROJECT_LOG.md`

This document explains, for anyone on the team, what we are doing, why, where every idea came from, what we found, and what is still open. It is updated at each real milestone (see the update log at the end).

1. The project in one paragraph

In many cities (e.g. Indian cities) piped water arrives only a few hours per day. A utility opens valves zone by zone. Households keep roof tanks to survive the gaps, and the utility cannot see how full those tanks are. We built a simulator (EPANET hydraulics via the WNTR package, network Net3, 4 zones) and trained reinforcement-learning agents to decide every hour which zones get water, aiming to serve as much demand as possible while protecting the worst-off zone (equity). We compare the agents with simple rules, an optimised fixed daily schedule, and (soon) a clairvoyant oracle.

2. How we got here (chronology)

1. Early October: the first project was SafeCharge-RL (EV charging). It is finished to a large extent and pushed on GitHub, but the faculty said EV charging is too common.
2. 30 Oct: we shelved it and searched for a new topic. An earlier Claude had written candidate files (`01-07`, about a *different* project: exam-aware study scheduling). I eliminated all 12 candidates against the other teams' topics (Excel sheet from the course).
3. 30-31 Oct: you brought `iws-rl.zip` (intermittent water supply), built by another Claude (Opus). We trained PPO, DQN and ablations and ran a full evaluation (v1 results).
4. 1 Nov: we found the v1 environment was too easy (see section 5), built the harder v2 environment, put everything on GitHub, and an external review (`docs/reference/improvements_for_ref.md`) sharpened the plan.
5. 1-2 Nov: three-laptop sprint (`docs/PLAN_v1.md`) to finish and present.

3. Where each idea came from

Idea / component	Source	Status
Topic: equity-aware RL for intermittent water supply (IWS)	Other Claude (Opus) produced `iws-rl.zip`; we chose it after eliminating 12 earlier candidates	Adopted, faculty approval to be confirmed (Adithyan)
EPANET hydraulics, network Net3	WNTR package (bundled example network), https://github.com/USEPA/WNTR	Tool
PPO, DQN, A2C implementations	Stable-Baselines3, https://github.com/DLR-RM/stable-baselines3	Tool
Household roof tanks, daily budget, equity reward	Original `iws-rl.zip` design; values (tank 0.3-0.8 days, alpha=4) are **assumptions**	Needs cited ranges (Adithyan)
Privileged greedy (reads hidden tank levels)	Added by us as an upper-ish reference	Not a true oracle (scores below PPO on v2)
Optimised fixed daily schedule (CE search)	Added by us as the stand-in for Bayesian-optimisation schedules in the literature (Hendrickson & Sela)	Under-tuned; mislabelled as BO in code; being redone
Diagnosis: scarce supply is degenerate	Our own analysis (`scarce_diag.py`, `scarce_trace.py`)	Done
Hydraulics never bind in v1; demand_scale=8 gives v2	Our own probes (`hyd_probe.py`, `scale_study.py`); the 4-8x peak-flow argument is our reasoning, not a citation	Done
Supply ceiling (mean supply can't exceed ~0.773 in v1)	External review `docs/reference/improvements_for_ref.md`; consistent with our scarce result 0.4754 vs ceiling 0.473	Consistent; not independently re-derived by us

Idea / component	Source	Status
Adaptive policy: domain randomisation + recurrent (LSTM) policy	Same review; resembles rapid-motor-adaptation ideas in robotics (from memory, **verify before citing**)	Planned (C1)
Equity as constraint / Pareto front; generalised Gini	Same review; Siddique et al., ICML 2020	Planned (C2)
Clairvoyant oracle for regret	Our design: env is deterministic given a seed, so optimise the full action sequence per scenario	Planned (Gate 1)

4. Related work found so far

Honesty note: every item below was read as a title or abstract/snippet only (web search). None has been read in full yet; full reading is task T3.1. Do not quote them as fact before that.

Closest to our problem (non-RL, equity in intermittent supply)

- Hendrickson & Sela, 2024: Bayesian optimisation of flow-control valve schedules for local supply and global equity, volume-driven demand with private tanks, 6 zones. https://doi.org/10.1016/j.scs.2024.105615 ; conference version https://www.mdpi.com/2673-4591/69/1/75 (our log earlier cited `69/1/184`, which may be wrong; to be checked)
- 2023 Water Supply, MINLP minimising supply inequity under hydraulic constraints: https://doaj.org/article/37953cf59e474f7687beb590577f3458
- Kurian et al. 2023, Water Supply, equitable supply in intermittently operated rural schemes: https://iwaponline.com/ws/article/23/11/4520/98117/Equitable-supply-in-intermittently-operated-rural
- UCSB intermittent-supply scheduling thesis: https://escholarship.org/uc/item/3xb4j737
- Bengaluru thesis on inequity under intermittent supply: https://etd.iisc.ac.in/handle/2005/5925
- Delhi/Bengaluru practice and equity discussion: https://www.indiawaterportal.org/health-and-sanitation/urban-sanitation/intermittent-water-distribution-networks-tale-two-cities

RL for water networks (no equity, not intermittent supply)

- Hu et al. 2020, deep RL valve scheduling for contamination isolation: https://aimspress.com/aimspress-data/mbe/2020/1/PDF/mbe-17-01-006.pdf
- Multi-agent RL (MADDPG) for pump and valve scheduling: https://bohrium.dp.tech/paper/arxiv/2408.08442
- PPO for pump scheduling on a real network with 18 demand zones: https://arxiv.org/pdf/2310.09412
- DRL-EPANET (NeurIPS workshop 2022): https://mlanthology.org/neuripsw/2022/belfadil2022neuripsw-drlepanet
- RL for household storage and tariffs in smart water networks (IFIP Networking 2025): https://dl.ifip.org/db/conf/networking/networking2025/1571132384.pdf
- Deep RL for pumps in water distribution systems: https://arxiv.org/pdf/2010.06460
- Safe RL control for water networks (Aalborg): https://vbn.aau.dk/en/publications/safe-reinforcement-learning-control-for-water-distribution-networ-2

Methods we may use (from the external review; not yet verified by us)

- Fair RL with generalised Gini welfare (Siddique et al., ICML 2020): https://arxiv.org/pdf/2008.07773
- Expected-welfare training does not guarantee fairness (two 2026 arXiv papers): https://arxiv.org/pdf/2606.18111 and https://arxiv.org/pdf/2608.00175
- Past-discounting for Markovian fairness: https://arxiv.org/abs/2504.01154v2
- Asymmetric actor-critic / teacher-student with privileged information (Cai et al., NeurIPS 2024): https://arxiv.org/abs/2412.00985
- CVaR in RL (Tamar et al.): https://arxiv.org/pdf/1406.3339v3.pdf ; Predictive CVaR policy gradient: https://rlj.cs.umass.edu/2024/papers/RLJ_RLC_2024_81.pdf

- Safe RL tooling: Safety-Gymnasium https://arxiv.org/pdf/2310.12567 ; OmniSafe https://github.com/PKU-Alignment/omnisafe
- Intermittent supply background (Dehra Dun): https://sswm.info/node/1445

Is it new?

Our searches (title/snippet level) found **no RL paper that combines intermittent supply, zone valve scheduling, equity and hidden household storage**. This is evidence of a gap, not proof. A proper audited novelty check is task T3.1. Also unconfirmed: a "Sept 2026 npj Clean Water equity paper" mentioned in an outside brief could not be found.

5. Results so far

v1 environment (150k-500k steps, results in `results/`*, summary* `results/RESULTS_SUMMARY.md`*)*

Method	mean supply	worst zone
PPO 500k (5 seeds)	0.767	0.737
PPO 150k (5 seeds)	0.766	0.735
Privileged greedy	0.766	0.736
PPO no equity / no forecast	0.763	0.724 / 0.727
Optimised fixed schedule	0.759	0.712
DQN	0.753	0.691
Greedy k=1	0.737	0.658
Random	0.701	0.618
A2C (untuned, failed)	0.440	0.143

What this means: v1 was **saturated**. Mean supply cannot exceed a ceiling of about 0.773 set by the daily water budget, and PPO reached 99% of it; the scarce-supply rows are an accounting identity (every model 0.4754 vs ceiling 0.473). Pressure thresholds of 10, 20, 30 m gave identical results because pressure stayed at 37-46 m (hydraulics never bound). PPO was also fragile: at surge factor alpha=2 it was worse than random.

v2 environment (demand_scale=8, tag `env-v2`*, results in* `results_v2/`*)*

Hydraulic probe: at demand x8, opening all zones delivers only 54-73% of requested flow (pressure 13-18 m), one zone 90-99%.

Method (default condition)	mean supply	worst zone
PPO (5 seeds, 300k)	0.716	0.601
Optimised fixed schedule (1 run)	0.708	0.612
Random	0.672	0.527
All open	0.648	0.513
Greedy k=2	0.648	0.589
Privileged greedy	0.639	0.494
Greedy k=1	0.604	0.591
Rota k=1	0.560	0.410

What this means: the sanity check passes (methods spread from 0.56 to 0.72; pressure thresholds now matter: 10 m 0.703, 20 m 0.716, 30 m 0.724). PPO beats greedy k=1 by +0.112 (CI 0.103-0.120). **But PPO is only about 0.008 ahead of the fixed schedule on mean supply and is worse on the worst zone**, so there is no clear RL win yet. The gap to the ceiling is about 0.057, which is the headroom an oracle may or may not be able to use (Gate 1). PPO-no-equity, DQN and A2C have not yet been run on v2.

Oracle (Gate 1) and the effect of domain randomisation (night of 1 Nov)

The **clairvoyant oracle** optimises the whole 72-step action sequence per test scenario with full knowledge of the future (hindsight upper bound, not a deployable policy). Its first version was invalid (it scored below PPO, so the search was too weak) and was replaced by a warm-started one.

Method (v2 default, 30 scenarios)	return (objective)	mean supply	worst zone
Oracle	-8.49	0.725	0.631
PPO	-9.48	0.719	0.601
Fixed schedule	-9.83	0.711	0.612

Gate 1 rule (>= 0.02 mean supply or >= 0.05 worst-zone headroom over the best non-PPO baseline) **fails** (0.013 and 0.019), but on the objective the oracle is +0.99 return over PPO: PPO closes only ~26% of the gap between the fixed schedule and the oracle. Little headroom on mean supply, some on worst-zone fairness.

Domain randomisation (C1). Plain PPO beats the fixed schedule in the normal setting (+0.38 return) but loses that advantage when the surge factor changes (alpha=2: -0.13, alpha=8: -0.02). PPO trained with randomised surge factor, tank sizes and budgets (3 seeds) beats the fixed schedule in every shifted condition (alpha=2 +0.39, alpha=8 +0.28, small tanks +1.03), at a cost of 0.2-0.4 return in-distribution. LSTM policies were **worse** than plain PPO (default return -11.2 and -13.9 vs -9.7), probably under-trained at 300k steps. Caveats: 3 seeds; one under-tuned fixed schedule; confidence intervals cover test scenarios, not training seeds.

Final M1 results (2 Nov morning): tuned fixed schedule, ablations, fairness Pareto

Source files: `results_v2/summary.csv`, `results_v2/paired_vs_ce_tuned.csv`, figures in `results_v2/figs/` (`pareto.png`, `regime_map.png`). Script: `analyze_final.py`. CIs are paired bootstraps over test scenarios only; training-seed variance is excluded. PPO has 5 seeds, ppo_dr/DQN/no-equity 3, tuned schedule 3 (CE 40 iterations, population 60), lambda sweep 2 per point.

Condition	PPO	PPO+DR	DQN	No-equity	Tuned schedule
default (return)	-9.67	-9.88	-10.04	-10.00	-9.15
alpha=2 (demand surge, shifted)	-9.67	-9.14	-8.31	-10.09	-10.81
small tanks (shifted)	-17.77	-17.20	-16.43	-18.50	-17.62
alpha=8	-11.18	-10.88	-11.88	-11.49	-10.98

- **The tuned fixed schedule beats plain PPO in the normal setting** (paired, significant), and in most conditions on both return and worst-zone supply. Our earlier "PPO beats the fixed schedule" was an artefact of an under-tuned baseline.
- **RL significantly beats the tuned schedule where conditions shift**: alpha=2 (all RL methods, return and worst-zone), small tanks (DQN and PPO+DR on return), scarce (return, DQN/PPO/PPO+DR). A fixed schedule cannot react to these.
- **Fairness weight lambda** (PPO, default): worst-zone 0.571 (lambda 0) to 0.635 (lambda 4) for a mean-supply cost of 0.010; node-level Gini does not improve. Removing the equity term lowers worst-zone supply (0.573 vs 0.601).
- **Honest limits:** tuned schedule overfits its training scenarios somewhat (train -7.4, test -9.15); single network; assumed parameters; the lambda points have 2 seeds; the novelty claim is unverified (literature audit pending).
- **Story:** RL is not better everywhere. It matters when the system differs from what a schedule was tuned for, and the equity weight gives a controllable fairness trade-off.

6. Open problems and the plan

- Fairness Pareto curve (C2) not finished: lambda sweep running on the MacBook, the Vivobook has the main version.
- No method contribution beyond environment + PPO -> C1 adaptive policy (domain randomisation + LSTM), C2 fairness Pareto front, C3 regime map (see `docs/PLAN_v1.md`).
- Fixed-schedule baseline is under-tuned and wrongly named (cross-entropy search, not Bayesian optimisation).
- No learning curves; single network; tank sizes and surge factor are assumptions; everything is simulation.
- Related work not yet read in full; faculty approval of the topic not yet confirmed.

7. Where things are in the repo

- `README.md` start here; `docs/PROJECT_LOG.md` chronological findings; `docs/PLAN_v1.md` the three-laptop plan; `docs/GIT_GUIDE.md` git for beginners; `docs/reference/improvements_for_ref.md` the external review.
- Code: `iws_env.py` (environment), `baselines.py`, `train_variant.py`, `eval_all.py`, `bo_schedule.py`, `v2cfg.py` (set `IWS_V2=1` for the harder environment), `run_jobs.py` (cross-platform job queue).
- Results: `results/` (v1), `results_v2/` (v2), models in `models/` and `models_v2/`.

8. Update log

Date	What changed
1 Nov 2026	Edition 1: project history, sources, v1 and v2 results, plan v1
night of 1 Nov	Edition 2: oracle and Gate 1, domain-randomisation and LSTM results

- 2 Nov morning (edition 3): added tuned-schedule baseline, no-equity/DQN runs, lambda Pareto, paired CIs; main conclusion changed (see section 5, final M1 results).